Risk-Aware Reinforcement Learning for Market Making Under Adverse Selection

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Abstract

Market making in modern electronic markets requires balancing profitability with exposure to adverse selection and inventory risk. While reinforcement learning (RL) has been proposed as a flexible framework for adaptive market making, its behavior under realistic microstructure constraints remains poorly understood. In this paper, we develop a high-fidelity limit order book simulator incorporating queue dynamics, stochastic order flow, and regime-dependent adverse selection. We evaluate a Soft Actor-Critic (SAC) agent against classical fixed-spread and imbalance-based skew strategies [4].

We show that naïvely trained RL agents converge to degenerate high-risk policies due to improper reward scaling, and that entropy regularization alone is insufficient to prevent this collapse. After correcting for reward scaling, the SAC agent learns a stable, risk-aware quoting strategy. Across multiple random seeds, SAC achieves comparable mean PnL to heuristic baselines while significantly reducing inventory exposure. Under increasing levels of adverse selection, SAC demonstrates superior robustness, maintaining positive conditional value-at-risk (CVaR) while baseline strategies suffer substantial tail losses.

These results suggest that RL-based market making is best viewed not as a profit-maximizing replacement for heuristics, but as a framework for adaptive risk management in adverse market conditions.

1. Introduction

Market making is a fundamental component of modern electronic financial markets, providing liquidity while earning the bid–ask spread. However, market makers face several key risks, including inventory exposure, adverse selection, and latency effects [2]. Designing strategies that balance profitability with risk control remains a central challenge.

Classical market making approaches, such as fixed-spread quoting or imbalance-based skewing, rely on hand-engineered rules and assumptions about market dynamics [1]. While these strategies are simple and interpretable, they lack adaptability to changing market regimes. Reinforcement learning (RL) offers a promising alternative by enabling agents to learn dynamic policies directly from interaction with a simulated environment. Prior work has explored RL-based market making, but often relies on simplified environments that omit critical microstructure features such as queue priority, stochastic order flow, and adverse selection [5].

Despite growing interest in RL for market making, existing work often relies on simplified environments that omit critical microstructure features such as queue priority, stochastic order flow, and adverse selection. As a result, RL agents may learn unrealistic strategies that exploit simulator artifacts rather than develop generalizable behaviors.

In this work, we address this gap by constructing a realistic, event-driven limit order book simulator with regime-dependent order flow and explicit modeling of adverse selection. Our formulation draws on the concept of order flow toxicity, where fills are followed by directional price pressure [3].

We train a Soft Actor-Critic (SAC) agent and compare its performance against standard heuristic baselines [4]. Our central finding is that RL's primary value in market making lies not in maximizing expected profitability, but in managing tail risk under adverse selection.

Specifically, while RL agents can initially exploit poorly scaled reward functions to achieve high apparent profitability, proper reward normalization leads to qualitatively different behavior. The resulting policies exhibit improved inventory control and robustness to adverse selection, though they do not consistently outperform heuristic strategies in raw PnL.

We make the following contributions:

1. We demonstrate that improper reward scaling leads to degenerate RL policies that exploit simulator dynamics, and that entropy regularization alone is insufficient to prevent this collapse.
2. We show that corrected training produces stable, risk-aware market making behavior with reduced inventory exposure.
3. We provide a systematic comparison between RL and heuristic strategies under varying levels of adverse selection.
4. We show that RL achieves superior tail-risk performance, as measured by CVaR, under stressed conditions—despite not maximizing mean PnL.

These results suggest that RL is most valuable not as a direct replacement for classical strategies, but as a tool for adaptive risk management in complex market environments.

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4. Methodology

4.1 Limit Order Book Environment

We construct an event-driven limit order book (LOB) simulator designed to capture key features of modern electronic markets while remaining computationally tractable [2]. The simulator maintains a central order book with price–time priority. Orders are represented as discrete price levels in tick space, and matching occurs whenever incoming orders cross the spread. The environment supports three primary order types: limit orders, market orders, and cancellations.Order flow is modeled as a stochastic point process. At each event, an order type is sampled according to regime-dependent probabilities, and quantities are drawn from geometric-like distributions. This produces a realistic mix of passive liquidity provision, aggressive trading, and order cancellations.

To capture changing market conditions, the simulator operates under a regime-switching process with distinct low-activity and high-activity states. Each regime specifies different arrival rates and order type distributions, allowing the market to exhibit periods of calm and heightened activity.

4.2 Adverse Selection Modeling

A key feature of the environment is the explicit modeling of adverse selection. When the agent's resting limit order is executed—that is, when the agent acts as a liquidity provider—subsequent order flow is biased against the agent [3].

Specifically, after a fill, the simulator probabilistically triggers a toxic flow regime for a fixed number of future events. During this period, market order arrival probabilities are shifted in the direction of the fill, and a directional drift is applied to order flow, increasing the likelihood of price movement against the agent's position. This mechanism ensures that providing liquidity exposes the agent to realistic selection risk, preventing the emergence of artificially profitable strategies based on symmetric or uninformed flow.

4.3 Agent Interaction

At discrete decision intervals, the agent observes a state vector derived from the current market and its internal position. The

observation includes the bid–ask spread (in ticks), the recent mid-price return, a rolling volatility estimate, top-of-book imbalance, normalized inventory, and time since last fill.

The agent submits a pair of limit orders (bid and ask) defined by price offsets from the mid-price and order sizes. All actions are bounded to prevent crossing the spread or submitting invalid orders. The agent's orders are treated identically to all other market participants, entering the same matching engine and subject to the same queue dynamics.

4.4 Reward Function

The agent's objective is defined using a mark-to-market profit and loss (PnL) formulation. At each step, the reward is computed as:

$$r_t = \Delta PnL_t - \lambda \cdot I_t^2 - \kappa \cdot Turnover_t$$

where ΔPnL_t is the change in mark-to-market portfolio value, I_t is the agent's inventory, λ controls inventory risk aversion, and κ penalizes excessive trading.

Critically, rewards are scaled by a constant factor to ensure stable learning. Without this normalization, the magnitude of PnL dominates both entropy regularization and risk penalties, causing the agent to learn degenerate policies that maximize short-term profit without regard for stability. We discuss this failure mode in detail in Section 7.1.

4.5 Reinforcement Learning Formulation

We model the problem as a Markov Decision Process (MDP) where the state consists of market features and agent inventory, the action specifies bid/ask quotes and sizes, the transition dynamics are governed by the stochastic LOB simulator, and the reward reflects risk-adjusted PnL [4].

We train a policy using the Soft Actor-Critic (SAC) algorithm, an off-policy actor–critic method with entropy regularization. The policy outputs continuous actions normalized to the range [−1, 1], which are then mapped to valid price offsets and order sizes. We employ automatic entropy tuning to adaptively balance exploration and exploitation during training.

4.6 Baseline Strategies

We compare the RL agent against two classical market making strategies:

- **Fixed Strategy (FIXED):** Quotes at a constant symmetric spread around the mid-price. This baseline represents a minimal, non-adaptive approach.
- **Skew Strategy (SKEW):** Adjusts quotes based on order book imbalance, biasing toward the side with higher expected short-term price movement. This baseline represents a commonly used directional heuristic.

Both baselines operate under the same market conditions and constraints as the RL agent.

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5. Experimental Setup

5.1 Training Configuration

We train the SAC agent with continuous action outputs. The policy is parameterized by a neural network mapping observations to action distributions over bid/ask offsets and sizes. Actions are normalized to [−1, 1] and mapped to discrete quote parameters within predefined bounds.

Reward scaling is applied to ensure that PnL magnitudes are comparable to entropy and penalty terms. Automatic entropy tuning adaptively adjusts the exploration coefficient during training; without this mechanism, we observe rapid policy collapse to extreme actions.

Each model is trained for 10,000 environment steps, with an initial warm-up phase of random actions to populate the replay buffer.

5.2 Evaluation Protocol

To assess robustness, we evaluate all strategies across multiple independent random seeds. We use a fixed training seed and 10 distinct evaluation seeds. Metrics are reported as mean and standard deviation across seeds. This multi-seed evaluation is critical due to the stochastic nature of order flow, which can produce misleading results from single-run evaluations.

5.3 Stress Testing via Adverse Selection

To evaluate performance under realistic market conditions, we introduce varying levels of adverse selection (toxicity), parameterized by the probability of triggering adverse flow after a fill, the duration of the toxic regime, the strength of directional bias in market orders, and the magnitude of directional drift [3].

We evaluate three regimes:

- **Light toxicity:** Short-lived, weak directional bias.
- **Medium toxicity:** Moderate duration and bias.
- **Strong toxicity:** Prolonged and strongly directional flow.

This setup allows us to analyze how strategies degrade as market conditions become increasingly adversarial—the central question of this paper.

5.4 Performance Metrics

We evaluate strategies using both profitability and risk-based metrics:

- **Mean PnL:** Average profit across evaluation runs.
- **PnL standard deviation:** Variability of returns.
- **CVaR (10%):** Average outcome in the worst 10% of runs.
- **Inventory range:** Maximum absolute inventory exposure during an episode.

While mean PnL captures overall profitability, CVaR and inventory range capture the tail-risk and risk management behavior that distinguish RL from heuristic approaches. Consistent with our thesis, we treat CVaR as the primary metric for evaluating strategy quality under adverse selection.

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6. Results

We evaluate the SAC agent against the FIXED and SKEW baselines. All results are averaged over 10 independent evaluation seeds.

6.1 Baseline Comparison

Under standard market conditions, SAC achieves intermediate mean PnL:

Policy	Mean PnL	Std
SAC	+6,635	4,856
FIXED	+5,033	3,049
SKEW	+8,988	6,192

SAC outperforms FIXED but does not exceed SKEW in raw profitability. However, SAC exhibits significantly improved inventory control, with an average inventory range of 7.9 compared to 14.2 (FIXED) and 12.1 (SKEW), as well as substantially lower variability in inventory exposure across seeds. These results indicate that SAC learns a more conservative policy, prioritizing risk control over aggressive profit maximization.

6.2 Toxicity Stress Test

Under increasing levels of adverse selection, a clear divergence emerges between strategies—particularly in tail-risk performance.

Light toxicity:

Policy	Mean PnL	CVaR (10%)

SAC	+10,188	+3,180
FIXED	+5,033	−1,354
SKEW	+8,988	+2,938

SAC achieves the highest mean PnL and maintains strong tail performance.

Medium toxicity:

Policy	CVaR (10%)
SAC	+1,194
FIXED	+322
SKEW	−3,910

SKEW performance deteriorates sharply in the tail. The directional signals that drive SKEW's profitability under benign conditions become liabilities when order flow is adversely selected.

Strong toxicity:

Policy	Mean PnL	Std	CVaR (10%)
SAC	8,970	3,420	+4,593
FIXED	4,987	4,323	−2,199
SKEW	5,222	3,386	+969

Under the most adverse conditions, SAC clearly dominates in tail-risk performance. While mean PnL remains comparable across strategies, SAC consistently avoids catastrophic losses. FIXED suffers from negative CVaR, indicating that its worst outcomes involve substantial losses. SKEW recovers somewhat in mean PnL but still exhibits weaker tail performance than SAC.

6.3 Summary of Findings

Three key findings emerge from these results:

1. **Reward scaling is critical.** Without normalization, RL agents converge to degenerate strategies that exploit simulator dynamics rather than learning meaningful risk–return tradeoffs.
2. **RL learns risk-aware behavior.** SAC reduces inventory exposure and avoids extreme positions, even without hard constraints on inventory.
3. **RL provides superior robustness under stress.** SAC significantly outperforms heuristic strategies in tail-risk metrics under adverse selection, even when it does not lead in mean PnL.

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7. Discussion

7.1 Reward Scaling and Policy Degeneracy

A central finding of this study is the critical role of reward scaling in RL for market making. When trained with unnormalized rewards, the SAC agent consistently converges to degenerate policies characterized by extreme actions and persistent boundary inventory positions. In this regime, the magnitude of mark-to-market PnL dominates both entropy regularization and risk penalties, effectively reducing the learning problem to short-term profit maximization without regard for stability.

Importantly, automatic entropy tuning alone is insufficient to prevent this collapse. Although the entropy coefficient increases significantly during training, the learned policy remains highly deterministic. This indicates that entropy regularization cannot compensate for poorly scaled value functions—a finding with practical implications for anyone applying RL to financial environments where reward magnitudes are inherently large.

After introducing reward normalization, the learned behavior changes qualitatively. The policy begins to utilize the interior of the action space, and inventory exposure is significantly reduced. This

demonstrates that proper reward scaling is not merely a training convenience, but a necessary condition for learning meaningful risk–return tradeoffs.

7.2 Emergence of Risk-Aware Policies

With appropriately scaled rewards, the SAC agent learns to manage inventory exposure without explicit constraints beyond a quadratic penalty term. Empirically, this manifests as reduced inventory range, lower variability in inventory across seeds, and avoidance of persistent directional positions.

This behavior suggests that the agent internalizes inventory risk as part of its optimization objective. Rather than maximizing raw PnL, the policy converges to a balanced strategy that trades off spread capture against inventory exposure. Notably, this behavior emerges from the interaction between the reward function and the environment dynamics rather than from hard-coded rules, highlighting RL's ability to discover nontrivial control policies in structured microstructure settings.

7.3 Sensitivity to Adverse Selection

The introduction of toxicity provides a controlled mechanism for studying adverse selection. Under increasing levels of toxicity, we observe a clear divergence between strategies. The SKEW strategy, which relies on directional signals derived from order book imbalance, exhibits large tail losses as adverse selection intensifies. This is because the very signals SKEW exploits—directional imbalance—become unreliable or misleading under toxic flow. The FIXED strategy maintains moderate performance but suffers from poor tail risk due to its inability to adapt. The SAC agent, by contrast, maintains consistently positive CVaR across all regimes.

This pattern suggests that RL policies are able to adapt dynamically to changes in order flow, particularly following unfavorable fills. In contrast, heuristic strategies apply fixed rules that cannot respond to regime-dependent risks—precisely the conditions where adaptive behavior provides the most value.

7.4 Tail Risk vs. Mean Performance

A recurring observation across all experiments is that SAC does not consistently outperform heuristic strategies in mean PnL under benign conditions. However, it significantly outperforms in tail-risk metrics under stress. This tradeoff is central to our thesis.

Heuristic strategies can achieve high average returns by exploiting stable patterns in order flow, but they are vulnerable to regime shifts and adverse selection. RL policies, by contrast, prioritize robustness and stability, sacrificing some upside in exchange for protection against catastrophic losses.

From a practical perspective, this suggests that RL-based market making is better suited for risk-sensitive applications—environments where avoiding large drawdowns matters more than maximizing average returns. This framing reorients the evaluation of RL in market making away from raw profitability and toward risk-adjusted resilience.

7.5 Implications for RL in Market Microstructure

These findings have broader implications for the application of RL in financial markets.

First, they demonstrate that realistic environment design is essential for RL in financial applications [5]. Without features such as adverse selection and proper reward scaling, RL agents may learn strategies that are unrealistic or uninformative—or worse, strategies that appear successful precisely because they exploit the simulator's limitations.

Second, they suggest that RL should be evaluated using risk-sensitive metrics such as CVaR, rather than mean returns alone. Mean PnL comparisons may obscure the adaptive advantages that RL provides under adversity.

Finally, they indicate that RL's primary advantage lies in its ability to adapt dynamically to complex and changing environments, rather than in outperforming well-designed heuristics in static or benign settings. This reframes the role of RL in quantitative finance: not as a replacement for domain expertise, but as a complement that provides robustness where fixed rules fail.

8. Conclusion

In this work, we study reinforcement learning for market making in a realistic limit order book environment with stochastic order flow and explicit modeling of adverse selection.

We show that naively trained RL agents converge to degenerate strategies that exploit reward scaling and simulator dynamics. Large unnormalized PnL signals lead to saturated policies that ignore inventory risk, and entropy regularization alone is insufficient to prevent this behavior. After introducing reward normalization, the learned policies exhibit qualitatively different behavior, including improved inventory control and reduced reliance on extreme actions.

Across multiple evaluation seeds, the SAC agent achieves competitive mean PnL relative to heuristic baselines while consistently maintaining lower inventory exposure. Under increasing levels of adverse selection, the RL agent demonstrates superior robustness as measured by CVaR, avoiding the large tail losses observed in both the fixed-spread and imbalance-based strategies.

These results support our central thesis: reinforcement learning is not necessarily advantageous for maximizing raw profitability in market making, but provides meaningful benefits in risk management and adaptability under adverse conditions. RL is most valuable not as a replacement for classical strategies, but as a framework for achieving resilience where heuristic approaches are most vulnerable.

Future work includes incorporating additional market frictions such as latency and queue position dynamics, as well as exploring formulations that directly optimize risk-sensitive objectives such as CVaR.

9. References

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